

LOYOLA
UNIVERSITY CHICAGO

COMP 271-03/400B-03: Data Structures I

Fall 2026 Syllabus

Tue/Thu, 1 pm – 2:15 pm CT @ 302 Cuneo Hall

Last Updated: Aug 19

Changelog

Please note that this syllabus may be subject to changes within the first few weeks of classes, mainly to enable minor corrections. Please check Sakai for any updates. Thank you for your understanding.

Course Information at a Glance

Course	COMP 271-03/400B-03: Data Structures I
Modality	In-person @ 302 Cuneo Hall
Duration	Aug 24, 2026 – Dec 12, 2026
Lecture Time	Tue/Thu, 1 pm – 2:15 pm CT
Instructor	Dr. Satyaki Sikdar
Instructor Email	ssikdar@luc.edu
Instructor Office	211 Doyle Hall
Office Hours	Online or in-person by appointment via Cal
Helpful Links	Office Hours Gradescope Late Pass Form

Course Components

Course Description

COMP 271 continues your development as a programmer, moving from the fundamentals covered in COMP 170 into the design, implementation, and analysis of data structures. Using Python, we will study how data can be organized, stored, and manipulated efficiently, and how the choice of a data structure shapes the performance and clarity of a program. The course also reinforces object-oriented design, since most data structures in this course are built and reasoned about as Python classes. The course emphasizes algorithm analysis, efficiency, and practical applications. You will also gain experience in testing and documenting code, writing software contracts, and using version control, preparing you for more advanced study in computer science and software development. Over the semester we will move from a review of core programming constructs and simple list operations into class design and Python's object model (special methods, class versus instance members), abstract data types, linked-list-based structures, software contracts and unit testing, stacks and queues, hash tables and hashing strategies, priority queues and heaps, and an introduction to graphs and graph traversal, closing with a review and technical-interview-style wrap-up.

Learning Outcomes

Upon successful completion of this course, you will be able to:

- **Data Structures**. Understand and implement fundamental data structures, including arrays, linked lists, stacks, queues, trees, and hash tables. Analyze their performance characteristics and choose appropriate structures for different applications.
- **Object-Oriented Programming**. Apply object-oriented programming principles such as encapsulation, inheritance, and polymorphism to design and implement modular, reusable code in Python.
- **Algorithm Analysis**. Evaluate the time and space complexity of algorithms using Big-O notation. Analyze the efficiency of common algorithms for searching, sorting, and manipulating data structures.
- **Practical Programming Skills**. Develop proficiency in Python programming, including testing and documenting code. Gain experience with version control systems and collaborative coding practices.
- **Problem Solving**. Formulate and solve computational problems using appropriate data structures and algorithms. Develop critical-thinking skills for approaching complex programming challenges.

Deliverables

Assignments

Assignments will be released weekly and will be due one week after they are released. The *lowest-scoring* problem set will be dropped when calculating the final grade. Over the semester, roughly 10 assignments will be released. Assignments will be administered on **Gradescope**.

Quizzes and Exams

There will be **4-5** timed in-person quizzes administered during class hours. There may be some overlap in content across quizzes. Quizzes may be written, oral, or a combination of both. They are intended to check individual understanding and may ask students to trace code, predict output, identify an error, explain a concept, or describe a solution. Make-up quizzes are provided only for documented emergencies, university-approved absences, approved religious accommodations, or other absences approved by the instructor.
The midterm and final exams may combine written questions, conceptual reasoning, code reading and tracing, and practical programming components. Their precise format, scope, location, duration, permitted resources, and technical requirements will be announced in advance on Sakai. Unless explicitly stated otherwise, quizzes and examinations are individual work, and the use of generative AI is **prohibited**.

Participation

Attendance is expected and recorded. Students may miss up to **three** class meetings without a reduction in attendance and participation grade. After the third absence, the participation grade will be reduced proportionally for each additional absence. University-approved absences, documented accommodations, and approved religious observances are handled according to University policy. Notify the instructor as soon as possible. Regardless of the reason, students must check Sakai, obtain class notes, and complete assigned work. Office hours don't replace missed class meetings. Participation includes arriving prepared, engaging in exercises, asking relevant questions, working constructively with classmates, and contributing to a respectful environment. Attendance alone doesn't guarantee full participation credit.

After each lecture, you will also be asked to write a brief **reflection essay** (minimum 3 sentences) covering the topics discussed. These will be administered on **Sakai** and will be due **24 hours** after the end of the lecture. These will be graded primarily for completion, and count towards your Participation grade. The **three lowest-scoring reflections** will be dropped while calculating the final grade.

Tools

Sakai

Sakai is the online course management system we will use to run the class. All course content, including the syllabus, lecture videos, assignments, solutions, and other course materials, will be posted here. It will also host your gradebook, which will be updated at least once a month. Please check the Sakai page at least once a week.

Gradescope

We will use Gradescope to administer the weekly problem sets. You can access Gradescope via the Assignments page on Sakai. For more information about Gradescope, please visit this website.

Microsoft Teams

The preferred way to contact the course staff is via direct messages on Microsoft Teams.

Schedule

The weekly schedule will be posted on the course Sakai page.

Requirements and Expectations

Course Textbooks

There is no mandatory textbook. Course materials, including lecture notes, assignments, and readings, will be provided online through the course website. Whenever readings are assigned, they will be drawn from accessible sources and posted alongside the corresponding topic.

What I expect of you

- Be fully present in class, abide by the Loyola University Honor Code, be kind to yourself, fellow students, and staff, and inform me promptly if you can't meet course obligations.
- Communicate professionally in all class-related communications.
- Complete assigned activities thoughtfully and on time.
- Enjoy learning about Data Structures.

What you can expect of me

- I provide well-organized and reasonable coursework to help you schedule your study time.
- I provide regular graded activities to track your progress.
- I offer advanced guidance on grading expectations.
- I grade work promptly and fairly.
- I provide necessary support and training for classroom success.

Evaluation and Grading

Grade Distribution

Component	***Weight***
Assignments	25%
Quizzes	30%
Midterm	15%
Final	20%
Participation	10%
Total	*100%*

The number, scope, and dates of individual assessments will be provided in the Sakai course schedule. The instructor may make reasonable adjustments to the assessment plan; changes will be announced in advance and will not disadvantage students.

Grade Scale

A	A–	B+	B	B–	C+	C	C–	D+	D	F
93+	90+	87+	83+	80+	77+	73+	70+	67+	60+	<60

Grade Change and Regrade Requests

Any requests for a grade change on an assignment or a quiz must be made within **seven** days when the grades are published. Any clerical errors while grading will be happily corrected. All change requests for problem sets must be made on **Gradescope**.

Course Policies

General

- Please be kind to yourself, your fellow students, and the course staff.
- A Jesuit education emphasizes learning to respect the rights and opinions of others. To do this, please respect others by (1) allowing all classmates to express their opinions without fear of ridicule and (2) refraining from using profanity or making objectionable (gendered, racial, or ethnic) comments, especially those directed at a classmate.
- Each instructor receives a class roster with the student's legal name. I'm happy to honor your request to address you by an alternate name and/or gender pronoun.
- Life can be unpredictable, and unexpected problems may arise that affect your life and work. If you need to address any issues related to this class, please contact Prof. Sikdar. If you have a problem with Prof. Sikdar, please contact the department chair, Prof. George Thiruvathukal.
- I'm committed to the principle of universal learning, which means that our classroom, virtual spaces, practices, and interactions are as inclusive as possible. Mutual respect, civility, and the ability to listen and observe others are essential for universal learning. Active, thoughtful, and respectful participation in all aspects of the course will make our time together productive and engaging.

Syllabus Changes

This syllabus communicates the intended structure and policies of the course. Changes may become necessary as the semester progresses. Any material change will be announced in advance in class and posted on Sakai. University policies in effect during the semester take precedence where applicable.

Attendance

The class will be conducted in-person. You are expected to attend every lecture and actively participate in class. Attendance will be taken at the start of every lecture.
At Loyola, faculty members will make efforts to accommodate students if the observance of a major religious holiday interferes with a student's academic work. If you are unable to attend a class, participate in a test or quiz, be present for a presentation, or complete an assignment on a specific day because of the observance of a religious holiday, you will be excused and provided the opportunity to make up the work. You will continue to be responsible for all assigned work and should notify me in advance through Loyola email of the religious observance(s) that conflict with your classes.

Late Work and Make Up

Life often gets in the way of study. To accommodate that, you have 2 Late Passes with no questions asked. **No late work will be accepted without late passes**.
To use a Late Pass, you must:

- Notify Prof. Sikdar before the assignment's due date via the late pass submission form. Your request will be approved automatically if you have late passes left.
- Submit the assignment within **3 calendar days** of the original due date, including a note that it's being submitted late with a Late Pass.
- Use only one Late Pass per assignment.
- Late Passes won't be applied automatically or retroactively to past assignments.

Academic Integrity

Academic integrity is the pursuit of scholarly activity in an open, honest, and responsible manner. Academic integrity is a guiding principle for all academic activity at Loyola University Chicago, and we expect all members of the University community to act under this principle. Failing to meet the following academic integrity standards seriously violates personal honesty and the academic ideals that bind the University into a learning community. These standards apply to both individual and group assignments.

Individuals working in a group may be held responsible if one of the group members has violated one or more of these standards.

1. Students may not plagiarize.
2. Students may not fabricate data.
3. Students may not collude or cheat.
4. Students may not facilitate academic misconduct.

Read through Loyola's full statement on Academic Integrity here. These policies fully apply to this course. The penalty for task-wise academic misconduct is losing all the task's points. Multiple events of misconduct will be reported through the proper channels.

Using Generative AI Tools

Gaining fluency in programming requires trial and error and sustained practice. This course aims to help students develop core problem-solving skills, computational thinking, and programming fluency. Students are discouraged from using generative AI as a primary means of obtaining solutions.

If a student uses generative AI to obtain *hints* or clarify concepts, they must **cite** the tool used and describe how the output was evaluated for correctness. The evaluation of an assignment may be affected by the extent to which AI was used to produce the work. In practice, this means generative AI tools may be used, when an assignment does not prohibit them, for hints, conceptual explanations, debugging guidance, or feedback on work you have already attempted. Any such use **must be disclosed** in the manner required by the assignment, including the tool used and enough information to understand how it influenced the submission.

Generative AI may **not** be used to generate a solution, or a substantial part of a solution, that is then submitted as your own work. You remain responsible for every line of code and every explanation you submit and must be able to explain, modify, and defend your work. Undisclosed or prohibited AI-generated work will be treated as an academic-integrity concern. When in doubt, ask before using a source, tool, or form of assistance if you are unsure whether it is permitted. Asking in advance protects both your learning and the integrity of your work.

Recording Notification

This course lectures may be recorded and distributed to you in a secure environment. You can watch these recordings on your computer, tablet, or smartphone. In Sakai, find the lecture video links on the course schedule. Your questions and comments may be recorded. Except for faculty and staff who require access, no content will be shared with others without your permission. Loyola University Chicago and your instructor jointly copyright these recordings. Posting them to other websites or without permission may result in disciplinary action and civil prosecution.

Privacy Statement

Recording online and face-to-face activities promotes open conversations and addresses concerns about sharing class comments. Faculty and students can use recordings solely for internal class purposes during the course. Recordings will be mentioned in the course syllabus. Instructors can use recordings with consent or remove student activity. They may retain recordings for personal use.

Intellectual Property

All course materials, including lectures, notes, and PowerPoints, are the professor's intellectual property. They may not be shared without written permission. Recordings of lectures require consent, which can only be used for review and not distributed. Similarly, I won't share or distribute your work without your permission.

Statement of Intent

By remaining in this course, students agree to accept this syllabus as a contract and to abide by the guidelines outlined in the document. Students will be consulted should there be a necessary change to the syllabus.

Notice of Reporting Obligations for Responsible Campus Partners

As an instructor, I am a Responsible Campus Partner (RCP) under Loyola's Comprehensive Policy and Procedures for Addressing Discrimination, Sexual Misconduct, and Retaliation. While my goal is for you to be able to engage fully and authentically with our course material through class discussions and written work, I also want to be transparent that as an RCP I am required to report certain disclosures of sexual misconduct (including sexual assault, sexual harassment, intimate partner and/or domestic violence, and/or stalking) to the Office for Equity & Compliance (OEC). As the University's Title IX office, the OEC coordinates the University's response to reports and complaints of sexual misconduct (as well as discrimination of any kind) to ensure students' rights are protected.
As an instructor, I also have an obligation under Illinois law to report disclosures of or suspected instances of child abuse or neglect.
The University maintains such reporting requirements to ensure that any student who experiences sexual/gender-based violence receives accurate information about available resources and support. Such reports will **not** generate a report to law enforcement (no student will ever be forced to file a report with the police). Additionally, the University's resources and support are available to all students even if a student chooses that they do not want any other action taken. If you have any questions about this policy, you are

encouraged to contact the OEC at equity@luc.edu or 773-508-7766.
If you ever wish to speak with a confidential resource regarding gender-based violence, I encourage you to call The Line at 773-494-3810. The Line is staffed by confidential advocates from 8:30am-5pm M-F and 24 hours on the weekend when school is in session. Advocates can provide support, talk through your options (medical, legal, LUC reporting, safety planning, etc.), and connect you with resources as needed – *without* generating a report or record with the OEC.

Student Accommodations

Loyola University provides reasonable accommodations for students with disabilities. Any student requesting accommodations related to a disability or other condition is required to register with Student Accessibility Center (SAC), located in Sullivan Center, Suite 117. Professors receive the accommodation notification from SAC via Accommodate. Students are encouraged to meet with their professor individually to discuss their accommodations. All information will remain confidential. Please note that software may be used to record class lectures to provide equal access to students with disabilities. Students approved for this accommodation use recordings for their personal study only and recordings may not be shared with other people or used in any way against the faculty member, other lecturers, or students whose classroom comments are recorded as part of the class activity. Recordings are deleted at the end of the semester. For more information about registering with SAC or questions about accommodations, please contact SAC at 773-508-3700 or SAC@luc.edu.

Mental Health

Each year, 5-20% of US residents contract the flu, while around 7% experience a major depressive episode. Depression is as common as the flu, yet one is socially stigmatized while the other isn't. As your professor, if mental health issues interfere with your ability to function, you can ask for short-term extensions or accommodations, just as you might for a physical illness. If you're struggling, please reach out. I'll work with you to minimize the impact on your performance.
If health problems, life stressors, or emotional difficulties interfere with your academic or personal success, contact the Wellness Center. They offer free, confidential healthcare services, crisis intervention, individual counseling, and group. You can make an appointment online or call 773-508-2530 for appointments or 773-508-8883 to speak with a nurse. If your medical or mental health condition requires ongoing academic accommodation, please register with Student Accessibility Services.